

Freight Rate Prediction Challenge

Machine Learning Engineer Assessment

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1. Introduction

The objective of this assessment was to build a machine learning model capable of predicting freight load rates using the provided historical shipment data. The development dataset was used to train and evaluate multiple regression models, after which the best-performing model was used to generate predictions for an unseen validation dataset and a fixed December forecasting dataset.

The solution emphasizes reproducibility, organized project structure, and a validation methodology designed to estimate how well the model generalizes to unseen freight loads.

2. Dataset

The project consisted of four primary datasets:

- `train-test.csv`
 - Historical freight records containing the target variable (rate) used for model training and evaluation.
- `validation.csv`
 - Unseen freight loads requiring predicted freight rates.
- `validation-predictions-template.csv`
 - Template used to produce the final prediction submission.
- `december_chart_inputs.csv`

- Fixed shipment characteristics for each day in December 2025 used to generate the required prediction chart.

Each shipment contains a combination of numerical and categorical features describing the load. The objective is to predict the freight rate for each shipment.

3. Data Preparation

Prior to model training, the data was preprocessed to improve model performance and ensure consistent feature representation.

The preprocessing pipeline included:

- Handling missing values through targeted imputation, equipment-specific medians for weight, and date-based medians for market signals, with global fallbacks for unseen categories or dates.
- Encoding categorical variables using one-hot encoding for equipment type and smoothed target encoding for lane, pickup, and delivery city.
- Constructing engineered features through a dedicated feature engineering module.
- Applying the identical preprocessing pipeline to both the validation dataset and the December prediction dataset to ensure consistency between training and inference.

Using a shared preprocessing pipeline prevents training-serving skew and guarantees that the model receives features in the same format during prediction as during training.

4. Model Selection

Four regression algorithms were evaluated using a held-out validation set:

Model	MAE	RMSE	MAPE
Ridge Regression	121.92	604.03	5.47%
Random Forest	150.64	615.28	6.68%
Gradient Boosting	151.59	618.31	7.00%
HistGradientBoosting	129.89	604.16	5.87%

Among the evaluated models, Ridge Regression achieved the lowest validation error across the selected metrics and was therefore chosen as the final model used for prediction.

5. Validation Strategy

The dataset was split temporally, using a fixed date cutoff, with earlier records used for training and later records held out for validation. This time-based approach better reflects the real-world task of forecasting future freight rates than a random split would.

Model performance was evaluated using three complementary regression metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

Evaluating multiple metrics provides a balanced assessment of model accuracy while reducing reliance on any single performance indicator. The same validation dataset was used when comparing all candidate models to ensure a fair comparison.

6. Prediction Generation

After selecting the best-performing model, predictions were generated for two required outputs:

1. The 12,000 freight loads contained in `validation.csv`, producing `validation_predictions.csv`.
2. The December forecasting dataset contained in `december_chart_inputs.csv`.

Both output files were successfully validated using the provided `score.py` utility before submission.

7. December Prediction Chart

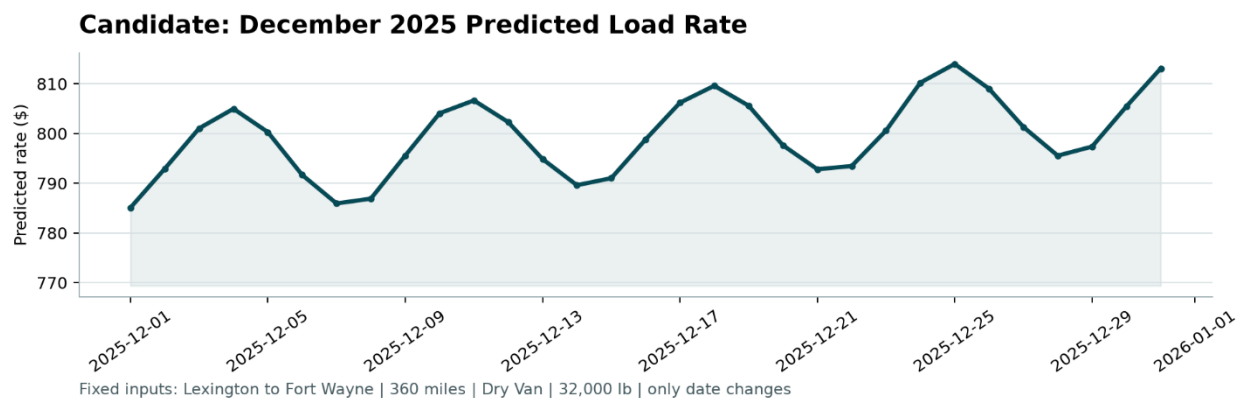


Figure 1: Predicted freight rates for the fixed Lexington–Fort Wayne shipment throughout December 2025.

8. Discussion

The evaluation results indicate that a regularized linear model generalized better than the more complex tree-based models on the provided dataset. This suggests that the relationship between the engineered features and freight rates is largely linear or that the regularization applied by Ridge Regression reduced overfitting.

Potential future improvements include:

- Additional feature engineering based on geographic relationships and seasonality.
- Hyperparameter optimization using cross-validation.
- Ensemble methods combining multiple regression models.
- Incorporating external freight market indicators if available.

Known limitations:

- Target encoding is in-fold. Lane/city smoothed means are fit and applied to the same training rows (not via out-of-fold cross-fitting). Smoothing (weight=20) limits this for low-count lanes, but it's a known simplification rather than a fully leakage-free encoding.
- No fallback for unseen cities. If a pickup/delivery city isn't in the coordinate lookup built from train/validation data, `feature_matrix()` will pass NaN coordinates straight into the model. Not an issue for the current

validation/December cities, but worth guarding against before running on new data.

9. Conclusion

A complete machine learning pipeline was developed to preprocess the data, engineer features, train multiple regression models, evaluate model performance, and generate predictions for both required datasets.

The final Ridge Regression model achieved the strongest validation performance among the evaluated models and successfully produced prediction files compatible with the provided scoring utility.